

MOHD AFSAR HUSSAIN

Complete Career Roadmap 2026

MERN Dev → DevOps / MLOps / LLMOps Engineer

Target: High-Paying Job in 8–12 Months

0-1 yr Current Exp

8-12 mo Timeline

**DevOps+MLOps
Target Role**

India+US Job Market

WHAT'S INSIDE THIS ROADMAP

- 1 Resume Gap Analysis — What's missing & what to fix NOW
- 2 Career Strategy — Why DevOps+MLOps is your best bet
- 3 Phase-by-Phase Learning Path — Month by month breakdown
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SECTION 1: RESUME GAP ANALYSIS

Tumhara current resume ek MERN Stack developer ka hai. Lekin tumhara target DevOps/MLOps hai — yeh ek MAJOR career pivot hai. Neeche analysis hai ki resume mein kya hai, kya missing hai, aur hiring managers kya dekhna chahte hain.

■ What's Already Good

- ✓ **Strong Academic Record:** CGPA 8.46 + 96% in Intermediate — yeh shortlisting mein help karta hai
- ✓ **MERN Stack Projects:** E-Cart + Weathy — real projects hain, JWT/Redis knowledge practical hai
- ✓ **Hackathon Winner:** Problem solving proof — interviewers ko yeh pasand aata hai
- ✓ **GitHub Active:** github.com/afsar-hussai — portfolio exist karta hai
- ✓ **Java + DSA Started:** 150 LC problems — yeh FAANG ke liye critical hai, sahi direction

■ Critical Gaps — Resume mein ye nahi hai (URGENT FIX)

Skill/Area	Status	Action Needed
Docker & Kubernetes	MISSING	Tum seekh chuke ho basics — ABHI resume mein add karo with a project
Ansible	MISSING	Oracle VM configuration project — yeh bhi resume mein add karo
GitHub Actions	MISSING	CI/CD pipeline — ek simple project ke saath show karo
Cloud (AWS/GCP/Azure)	NOT MENTIONED	Koi ek cloud seekhna mandatory hai — AWS recommend hai
MLOps Tools	NOT MENTIONED	MLflow, DVC, kubeflow — targeted roles ke liye must hai
Monitoring Tools	NOT MENTIONED	Prometheus, Grafana — DevOps JD mein 80% mein listed hai
Python	NOT MENTIONED	MLOps/DevOps automation ke liye mandatory, tumhara Java background help karega
Professional Summary	WRONG FOCUS	'Full-Stack Web Developer' likhna band karo — DevOps Engineer target karo

■ IMMEDIATE ACTION (This Week): Apna Professional Summary change karo to: "DevOps & MLOps Engineer with hands-on experience in Docker, Kubernetes, Ansible, GitHub Actions CI/CD pipelines. MERN stack background with strong problem-solving skills. Passionate about building scalable infrastructure and ML deployment pipelines."

SECTION 2: CAREER STRATEGY

Why DevOps + MLOps? Tumhare liye Best Bet kyun hai?

Tumhara MERN background actually ek ADVANTAGE hai DevOps mein — tumhe already APIs, deployment concepts, aur Node.js pata hai. Yeh combination bahut rare hai aur companies prefer karti hain engineers jo full stack + infra dono samjhein.

Path	Target Salary (India)	Target Salary (US Remote)	Timeline	Difficulty
DevOps Engineer (Mid Level)	■15–30 LPA	\$80K–\$120K	6–8 months	Medium
MLOps Engineer	■20–50 LPA	\$120K–\$160K	8–10 months	Medium-High
LLMOps Engineer	■40–100 LPA	\$150K–\$200K+	10–14 months	High
Platform Engineer	■25–55 LPA	\$130K–\$180K	10–12 months	High

Recommended Primary Path for You:

DevOps Engineer (Month 1–6) → MLOps Engineer (Month 7–10) → LLMOps Specialist (Month 10–12)
Reason: Tumhare paas Docker, K8s, Ansible already basics hain. Pehle DevOps solid karo, ek job lo mid-way ya end mein. Phir MLOps/LLMOps add karo for max salary. Yeh progressive approach risk bhi kam karta hai aur salary bhi zyada.

SECTION 3: PHASE-BY-PHASE LEARNING PATH

PHASE 1

Foundation Solidify + Resume Fix · Month 1–2

■ Goal: Resume job-ready banana + DevOps basics strong karna

Topic	What to Learn & Why
Linux & Bash Scripting	File permissions, cron jobs, shell scripts — DevOps ka foundation. Agar weak hai toh yahan se shuru karo.
Docker Deep Dive	Tumhe basics pata hai, ab advance karo: multi-stage builds, Docker Compose, networking, volumes, image optimization.
Kubernetes (K8s)	Pods, Deployments, Services, ConfigMaps, Secrets, Ingress. Minikube/kind se local practice. CKA prep shuru karo.
Git & GitHub Advanced	Branching strategies, GitHub Actions CI/CD pipelines. Ek real pipeline banao apne MERN project ke liye.
Networking Basics	TCP/IP, DNS, Load Balancers, Reverse Proxy (Nginx) — interviews mein yeh zaroor aate hain.

■ Projects to Build in This Phase

- Dockerize tumhara E-Card MERN project with multi-stage builds
- Deploy it on Kubernetes (minikube) with proper services/ingress
- Setup GitHub Actions CI/CD pipeline for auto-build + deploy
- Ansible playbook for server configuration (extend Oracle VM project)

■ Phase Deliverable: Updated resume with Docker, K8s, Ansible, GitHub Actions + 2 solid projects on GitHub

PHASE 2

Cloud + Monitoring + IaC · Month 3–4

■ Goal: Cloud skills add karna + Observability + Infrastructure as Code

Topic	What to Learn & Why
AWS Core Services	EC2, S3, VPC, IAM, RDS, EKS, ECR, CloudWatch. AWS Free Tier use karo for hands-on. AWS SA Associate cert target karo.
Terraform (IaC)	Write infrastructure as code. Deploy AWS resources via Terraform. State management, modules, workspaces.
Prometheus + Grafana	Setup monitoring stack, write PromQL queries, create dashboards. Must-have for SRE/DevOps JDs.
Helm Charts	Package K8s applications with Helm. Write custom charts. Used everywhere in production K8s setups.
Python for DevOps	Boto3 (AWS SDK), scripting, automation. 2-3 hours/week enough. Already Java jaante ho so easy transition.

■ Projects to Build in This Phase

- Deploy E-Card on AWS EKS using Terraform (full IaC)
- Setup Prometheus + Grafana monitoring for your K8s cluster
- Write Helm charts for your application
- Python script: Auto-scale EC2 instances based on CPU metrics

■ **Phase Deliverable: AWS hands-on experience + Terraform IaC project + Monitoring dashboard — ready for DevOps job applications**

PHASE 3

MLOps Core Skills · Month 5–7

■ **Goal: MLOps engineer ke roop mein position karna**

Topic	What to Learn & Why
ML Fundamentals (Light)	Tumhe ML researcher nahi banna — sirf basics: linear regression, classification, what is a model, train/test split. Scikit-learn use karo.
MLflow	Experiment tracking, model registry, model versioning. Sab se important MLOps tool. Seedha seekho.
Kubeflow / ZenML	ML pipeline orchestration on Kubernetes. Production-grade ML workflows.
DVC (Data Version Control)	Data aur model versioning — MLOps mein Git jaisi role hai iski.
Feature Stores & Model Serving	FastAPI se model serve karna, BentoML, basic Feast knowledge.
Airflow / Prefect	Pipeline orchestration — data engineering ka core, MLOps mein bhi use hota hai.

■ Projects to Build in This Phase

- End-to-end ML pipeline: data ingestion → training → MLflow tracking → deployment as FastAPI
- Kubeflow pipeline on local K8s cluster for a classification model
- Model drift monitoring with Evidently AI
- DVC + Git setup for dataset versioning

■ **Phase Deliverable: MLOps portfolio project + LinkedIn headline update to 'DevOps & MLOps Engineer'**

PHASE 4

LLMOps + Advanced Projects + Job Hunt · Month 8–12

■ **Goal: LLMOps skills + aggressive job applications + FAANG prep**

Topic	What to Learn & Why
LangChain / LangGraph	LLM orchestration, chains, agents. Tumhare DevOps background se yeh directly applicable hai production mein.
Vector Databases	Pinecone, Weaviate, ChromaDB — RAG applications ke liye must. Ek RAG chatbot banao.
vLLM / TensorRT-LLM	High-performance LLM inference — GPU ops, model serving at scale. High-salary niche.

LangFuse / LangSmith	LLM observability — prompt versioning, trace analysis, cost monitoring.
DevSecOps	Trivy, Falco, OPA/Gatekeeper — add karo portfolio mein for 20-35% salary boost.
System Design	Design distributed systems, databases at scale, microservices. FAANG rounds mein critical.

■ Projects to Build in This Phase

- RAG-based chatbot: LangChain + Pinecone + FastAPI + Docker + K8s deployment
- LLM inference server with vLLM — monitor with LangFuse
- Full MLOps + LLMOps pipeline on AWS EKS (capstone project)
- Open source contribution: MLflow, LangChain, or Kubeflow

■ Phase Deliverable: High-salary JD fit resume + 4-5 strong GitHub projects + actively applying to 10+ jobs/week

SECTION 4: WEEKLY TIMETABLE

Tumhari shift: 11 AM – 8 PM (Mon–Fri) | Saturday–Sunday OFF Available study time: Morning before shift + After shift + Full weekends

Weekday Schedule (Mon–Fri)

Time Slot	Activity	Duration	Focus
6:00 – 7:30 AM	DSA Practice (LeetCode)	90 min	2 problems/day — patterns focus
7:30 – 8:00 AM	Morning routine + commute prep	30 min	—
8:00 – 9:00 AM	Theory Learning (Video/Docs)	60 min	Current phase topic — no distractions
9:00 – 10:30 AM	Hands-on Practice / Lab	90 min	Docker / K8s / Terraform / MLflow etc.
10:30 – 11:00 AM	Commute / Get ready	30 min	Podcast/Tech talk (optional)
11:00 AM – 8:00 PM	WORK SHIFT	—	Focus on job. Lunch break: revise notes
8:00 – 8:30 PM	Dinner + Short break	30 min	Rest — no screens
8:30 – 9:30 PM	Project building / Implementation	60 min	Build the phase project
9:30 – 10:00 PM	GitHub commit + Notes update	30 min	Daily GitHub commit habit
10:00 – 10:30 PM	System Design / LLMOps concepts	30 min	Light reading before sleep
10:30 PM	SLEEP	—	8 hrs sleep = better learning retention

Weekend Schedule (Sat–Sun) — Power Learning Days

Time Slot	Activity	Duration
7:00 – 9:00 AM	DSA — 3 medium/hard problems + review patterns	2 hrs
9:00 – 10:00 AM	Breakfast + Break	1 hr
10:00 AM – 1:00 PM	DEEP WORK SESSION 1 — Major topic learning + lab	3 hrs
1:00 – 2:00 PM	Lunch + Rest	1 hr
2:00 – 5:00 PM	DEEP WORK SESSION 2 — Project building / deployment	3 hrs
5:00 – 6:00 PM	Break / Exercise / Cricket	1 hr
6:00 – 8:00 PM	System Design study / Mock interview prep	2 hrs
8:00 – 9:00 PM	Weekly review: What learned? What to fix next week?	1 hr
9:00 PM	Relax — music, cricket, free time	—

Weekday Study ~4.5 hrs/day	Weekend Study ~10 hrs/day	Weekly Total ~42–43 hrs	This is SERIOUS preperation pace
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SECTION 5: DSA PLAN — FAANG CRACK STRATEGY

Tumne 150 LC problems solve ki hain aur sab DSA concepts clear hain — yeh bahut achha hai. Ab sirf patterns aur problem-solving speed improve karna hai.

Pattern-Based Approach (Most Important):

Pattern	Difficulty	Problems to Solve
Sliding Window	Easy→Medium	Subarray problems, max/min window — 8–10 problems
Two Pointers	Easy→Medium	Sorted arrays, palindromes, pair sum — 8–10 problems
Fast & Slow Pointers	Medium	Linked list cycle, find middle — 6–8 problems
Binary Search	Medium→Hard	Search in rotated array, min in mountain — 10–12 problems
BFS/DFS (Graphs)	Medium→Hard	Islands, shortest path, clone graph — 12–15 problems
Dynamic Programming	Hard	Knapsack, LCS, coin change — 15–20 problems. Most asked in FAANG
Trees (DFS/BFS)	Medium→Hard	Level order, diameter, LCA — 12–15 problems
Heaps & Priority Queue	Medium	Kth largest, merge K lists — 8–10 problems
Backtracking	Medium→Hard	Permutations, subsets, N-queens — 8–10 problems
Monotonic Stack	Medium→Hard	Next greater element, histogram — 6–8 problems

Month-by-Month DSA Target:

Month 1–2	Complete all 9 patterns above. Solve 2 problems/day on weekdays, 3/day on weekends. Total: ~130 new problems.
Month 3–4	LeetCode company-wise problems: Amazon, Google, Microsoft (filter on LC). Focus on Medium-Hard. Mock interviews on Pramp.com 1x/week.
Month 5–6	Advanced DP + Graph algorithms. Contest participation: Codeforces/LC weekly. Goal: 400 total LC problems solved.
Month 7–8	System Design ko DSA ke saath parallel chalao. Mock interviews on Interviewing.io. 2 mocks/week.
Month 9–12	Full mock interview rounds (DSA + System Design + DevOps concepts). Target: 500+ LC problems, consistent solve speed under 20 min for medium.

SECTION 6: CERTIFICATIONS ROADMAP

Certifications tumhara resume strong banate hain, especially jab experience kam ho. Order matters — yeh sequence follow karo.

#	Certification	When	Cost	Priority	Why Take It
1 s t	CKA — Certified Kubernetes Administrator	Month 2–3	~\$395	HIGH	Most valued DevOps cert. K8s seekhte hi exam do. Almost every product company JD mein listed.
2 n d	AWS Solutions Architect Associate	Month 3–4	~\$150	HIGH	20–25% salary boost proven. Free tier practice + TutorialsDojo mock tests. Sab se popular cert globally.
3 r d	HashiCorp Terraform Associate	Month 4	~\$70	MEDIUM	IaC skills validate karta hai. AWS cert ke baad seedha yeh lo — content overlap hai.
4 t h	AWS DevOps Engineer Professional	Month 6–7	~\$300	HIGH	Senior-level cert. CI/CD, monitoring, IaC sab cover. US remote jobs ke liye strong signal.
5 t h	CKAD — Certified K8s Application Developer	Month 5	~\$395	MEDIUM	CKA ke baad lo. MLOps + K8s combination ke liye useful.
6 t h *	Google Cloud Professional ML Engineer	Month 9–10	~\$200	HIGH for MLOps	MLOps role ke liye sab se relevant cert. GCP Vertex AI + ML pipeline knowledge validate karta hai.

* Note: CKA exam \$395 hai —agar budget tight hai toh pehle AWS SA Associate (\$150) lo, usse job lo, phir CKA lo company reimbursement mein (most companies cover cert costs).

SECTION 7: BEST RESOURCES — FREE & PAID

Linux & Bash

The Linux Command Line (book)	FREE — overthewire.org bandit wargame for practice
Ryan's Linux Tutorial	FREE — ryanstutorials.net

Docker & Kubernetes

KodeKloud	PAID (~\$20/mo) — Best hands-on labs for CKA prep. Worth every rupee.
TechWorld with Nana (YouTube)	FREE — Best K8s + Docker playlist. Watch in order.
Killer.sh	PAID — CKA exam simulator. Use 2 weeks before exam.

Terraform & IaC

HashiCorp Learn (learn.hashicorp.com)	FREE — Official tutorials, best quality
Anton Putra (YouTube)	FREE — Terraform + AWS excellent tutorials

AWS

TutorialsDojo Practice Exams	PAID (~\$15) — Best AWS mock tests for SA Associate
Adrian Cantrill's Course (learn.cantrill.io)	PAID (~\$40) — Best in-depth AWS course
AWS Free Tier + Skill Builder	FREE — Official AWS hands-on labs

MLOps

MLOps.community (YouTube + Discord)	FREE — Real practitioners, best content
Made With ML (madewithml.com)	FREE — Best MLOps practical course online
Weights & Biases Courses	FREE — MLflow, W&B, experiment tracking
Full Stack Deep Learning (fullstackdeeplearning.com)	FREE — Production ML systems

LLMOps / LangChain

LangChain Documentation + Cookbook	FREE — Official docs + examples
DeepLearning.AI Short Courses (deeplearning.ai)	FREE — LangChain, LLMOps, RAG courses
LlamaIndex Docs	FREE — Alternative to LangChain, learn both

DSA

NeetCode.io	FREE — Best organized LC patterns, video solutions
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LeetCode Premium	PAID (~\$35/mo) — Company-specific problems worth it Month 4+
Pramp.com	FREE — Mock interviews with real engineers
Interviewing.io	FREE trial — Anonymous mock interviews

System Design

System Design Primer (GitHub)	FREE — github.com/donnemartin/system-design-primer
Alex Xu's System Design Interview (book)	PAID (~\$30) — Must read for FAANG
Gaurav Sen (YouTube)	FREE — Best Hindi/English system design channel

Monitoring

Prometheus + Grafana Official Docs	FREE
TechWorld with Nana — Prometheus playlist	FREE YouTube
Datadog Learning Center	FREE — learn.datadoghq.com

SECTION 8: JOB APPLICATION STRATEGY

When to Start Applying?

Phase	When to Apply	What to Target	Expected Package
Phase 2 end (Month 4)	Start sending applications	Junior DevOps/Infrastructure roles Service-based: TCS, Infosys, Wipro, HCL (DevOps vertical)	India: ■8–15 LPA US: Not yet
Phase 3 mid (Month 6–7)	Aggressive applications	DevOps Engineer (Product companies) Startups: Razorpay, CRED, Zepto, Meesho US remote: Junior DevOps	India: ■15–25 LPA US: \$60–90K remote
Phase 4 (Month 9–12)	Full-force — FAANG + mid-level	Mid-level DevOps/LLMOps roles FANG-India: Microsoft, Google, Amazon US remote: Mid-level	India: ■25–50 LPA US: \$100–160K

LinkedIn + Resume Strategy

- Headline: 'DevOps & MLOps Engineer | Docker | Kubernetes | AWS | GitHub Actions' — change this WEEK 1
- Post weekly: 1 LinkedIn post about what you learned (builds visibility, recruiters notice)
- GitHub: Daily commits — even small ones. Green contribution graph = credibility
- Follow & engage: Kelsey Hightower, Liz Rice, Chip Huyen (MLOps) — learn + get visible
- Open to Work: Turn on, set to 'DevOps Engineer / MLOps Engineer / Cloud Engineer'
- Connections: Add 5 DevOps engineers/recruiters per week on LinkedIn with personalized note
- Apply on: LinkedIn Easy Apply, Naukri.com (India), lever.co, greenhouse.io (US), Wellfound (startups)
- US Remote specifically: Turing.com, Toptal, Deel.com, Remote.co, WeWorkRemotely.com

Interview Preparation Checklist

Round Type	Focus Areas
DevOps Technical Round	Docker networking, K8s troubleshooting, CI/CD debugging, Terraform state issues, monitoring queries
MLOps Technical Round	Model deployment pipeline, experiment tracking, drift detection, feature stores, model serving
DSA Round (FAANG)	2 medium LeetCode in 45 min. Focus: arrays, trees, graphs, DP
System Design Round	Design Netflix, Uber, WhatsApp scale systems. Practice distributed systems concepts
Behavioral Round (STAR)	Situation, Task, Action, Result — prepare 5-6 stories from your projects + hackathon win
Cloud Architecture Round	Design scalable AWS infrastructure, cost optimization, high availability patterns

SECTION 9: 12-MONTH MILESTONE TRACKER

Month	Key Milestone & Deliverable
Month 1	Linux solid, Docker advanced, K8s basics, GitHub Actions CI/CD pipeline live, Resume updated, 60 new LC problems, Dockerized E-Card on GitHub
Month 2	K8s advanced (CKA prep), Ansible playbook expanded, Helm charts, 50 LC problems, Mock K8s interview, CKA exam booked
Month 3	CKA CERTIFIED, AWS basics started (Free tier), Terraform basics, Prometheus+Grafana setup, 50 LC problems, First DevOps job applications sent
Month 4	AWS SA Associate CERTIFIED, Full IaC project on AWS+Terraform+K8s, Terraform cert, 50 LC problems, 10 applications/week
Month 5	Python basics for DevOps, CKAD certified, MLflow experiment tracking, DVC setup, 40 LC problems, System Design study started
Month 6	Kubeflow pipeline project, FastAPI model serving, Airflow DAG, 40 LC problems, 10-15 applications/week, Mock interviews started
Month 7	MLOps capstone project complete, AWS DevOps Pro exam prep, Data drift monitoring (Evidently AI), 40 LC problems, Active job hunt
Month 8	AWS DevOps Engineer Professional CERTIFIED, LangChain basics, vector DB intro, 30 LC, System design mock 2x/week, Interviews happening
Month 9	RAG chatbot project built + deployed on K8s, LangFuse monitoring, 30 LC, GCP ML cert prep, Targeting LLMOps roles
Month 10	GCP Professional ML Engineer CERTIFIED, vLLM inference server, DevSecOps tools added, 30 LC, Salary negotiation practice
Month 11	Open source contribution (MLflow/LangChain), Full capstone project documented, 500+ LC total, Final interview prep rounds
Month 12	TARGET: JOB OFFER RECEIVED. DevOps/MLOps/LLMOps role. India: ₹20–50 LPA / US Remote: \$100K–160K

FINAL NOTE: Bhai, tumhara combination — MERN background + DevOps basics + MLOps interest + DSA preparation — yeh 2026 ka perfect storm hai. Market mein log akele DevOps jaante hain, ya akele ML jaante hain. Tum dono seekh rahe ho. Yeh rare hai. Consistency is everything. Ek din miss hua toh problem nahi — 1 week miss hua toh recover karna mushkil hota hai. Is roadmap ko daily dekho. GitHub ko daily commit karo. Aur remember: ek saal ki mehnat = decades ki financial freedom. All the best, Afsar! ■